

1. Give a relation about dentist/patient appointment data. A patient is given an appointment at a specific time and date with a dentist located at a particular surgery. On each day of patient appointments a dentist is allocated to a specific surgery for that day.

Appointment (staffNo, dentistName, patNo, patName, date, time, surgery)

(a) Identify all the functional dependencies within the above relation; (point: 2)

staffNo $\rightarrow$ dentistName

patNo $\rightarrow$ patName

staffNo, date $\rightarrow$ surgery

patNo, date, time $\rightarrow$ staffNo, dentistName, patName, surgery

patNo, staffNo, date $\rightarrow$ dentistName, patName, surgery, time

staffNo, date, time $\rightarrow$ patNo, patName, dentistName, surgery

surgery, date, time $\rightarrow$ U

(b) Identify the primary key of this table; (point:2)

Pk: patNo, staffNo, date

CK:

(c) What normal form is the relation in? Explain your answer. (point: 2)

1NF, not 2NF, because there are some nonprimary key attributes that are partially dependant on the PK

(d) Apply normalization until the relations are BCNF. State the reasons behind each decomposition. (point:6)

(1) 1NF $\rightarrow$ 2NF

Appointment' (staffNo, patNo, date, time)

PK (staffNo, patNo, date)

Dentist(staffNo, dentistName)

PK: staffNo

Patient(patNo, patName)

PK: patNo

surgery(staffNo, date, surgery)

PK: staffNo, date

(2) 2NF $\rightarrow$ 3NF

The above relations schema are all 2nf

(3) 3NF $\rightarrow$ BCNF

All determinants are PK, so they are all BCNF

staffNo, date, time $\rightarrow$ patNo, patName, dentistName, surgery

Appointment' (staffNo, patNo, patName, date, time)

PK: staffNo, date, time

Dentist (staffNo, dentistName)
Surgery(staffNo, date,surgery)

Appointment' (staffNo, patNo, patName, date, time)
PK: staffNo, date,time
Is 2nf
Is not 3nf

Appointment'' (staffNo, patNo, date, time)
PK: staffNo, date,time
staffNo, date,time→patNo
patNo→ patName
so patName is transitively dependant on the PK
Patient(patNo,patName)

2. Consider the following relations that represent the information of an invoice of a product sale:

Invoice_data(I_number, I_date, C_number, C_name, C_city, Item_number, Item_name, Item_qty, Item_price, Total price).

Each invoice (发票) has a number(I_number), date(I_date), a single customer number (C_number), total price(Total price) of all items.

Each invoice may list one or more items that the customer bought, and each item in the invoice has a unique item number(Item_number), name of item(Item_name), quantity(Ttem_qty) and price(Item_price).

Items of product may be bought by more than one customer.

Each customer has a customer number (C-number), customer name(C_name), customer city(C-city).

(a) Identify all the functional dependencies within the above relation; (point: 2)

I_number→I_date, C_number, C_name, C_city, Total price

Item_number→ Item_name, Item_price

I_number, Item_number→u

C_number→C_name, C_city

(b) Identify the primary key of this table; (point:2)

PK: I_number, Item_number

(c) What normal form is the relation in? Explain your answer.(point: 2)

1NF, NOT 2NF , there are some nonprimary key attributes that are partially dependant on the PK

(d) Apply normalization until the relations are BCNF. State the reasons behind each decomposition. (point:6)

1NF→2nf

Invoice_data'(I_number, Item_number, Item_qty).

I_number, Item_number → Item_qty

Invoice_S(I_number, I_date, C_number, C_name, C_city, Total price)

Product(Item_number, Item_name, Item_price)

2NF → 3NF

Invoice_data'(I_number, Item_number, Item_qty).

I_number, Item_number → Item_qty

It is 3NF and it is also BCNF

Product(Item_number, Item_name, Item_price)

PK: Item_number

Item_number → Item_name, Item_price

It is 3NF and it is also BCNF

Invoice_S(I_number, I_date, C_number, C_name, C_city, Total price)

I_number → I_date, C_number, C_name, C_city, Total price

C_number → C_name, C_city

C_name, C_city are transitively dependant on PK

So it is no 3NF

Invoice_S'(I_number, I_date, C_number, Total price)

I_number → I_date, C_number, Total price

Is BCNF

Customer(C_number, C_name, C_city)

C_number → C_name, C_city

The PK C_number is the only determinant, So the schema is BCNF